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# GEOGRAPHY · Interior of the Earth: rocks, earthquakes and volcanoes

**UNDERSTAND IT FIRST**

**What is it?** Nobody can travel to the centre of the Earth (the deepest hole ever drilled, the **Kola borehole** in Russia, is only about **12 km** deep, while the centre is **6,371 km** down). So scientists study the interior **indirectly**, mainly through **earthquake (seismic) waves**. The interior explains **why the ground shakes, why volcanoes erupt, how mountains and oceans form and what rocks we stand on**. Learn it as **how we know → seismic waves → layers → rocks → moving plates → earthquakes → volcanoes**.

## 1. How we know: sources and seismic waves

- **Direct sources:** mines and deep drilling (only a few km), and **lava** from volcanoes. **Indirect sources: seismic waves**, rise of **temperature and pressure** with depth (about **1°C for every 32 m** near the surface), **gravity** and **magnetism** differences, and **meteorites** (thought to be made of material like the Earth's interior).
- **Body waves: P (primary) waves** are the **fastest**, travel like **sound** (push–pull) and pass through **solids, liquids and gases**. **S (secondary) waves** move the ground **up and down (sideways)** and pass **only through solids**. **Surface waves** (Love and Rayleigh) are the **slowest** but cause the **most damage**. Between about **105° and 145°** from the epicentre lies a **shadow zone** with no direct P or S waves; beyond **105°**, **no S waves** arrive at all, proving the **outer core is liquid**.

Seismic waves and the shadow zone

## 2. Layers of the Earth

Layers of the Earth

| Layer | Facts |
|---|---|
| **Crust** | Thin outer skin: about **5 km** under oceans, about **30–35 km** on average under continents and up to **70 km** under big mountains. **Continental crust** is lighter, **granitic**, rich in silica and aluminium (**SiAl**); **oceanic crust** is heavier, **basaltic**, rich in silica and magnesium (**SiMa**). The **Conrad discontinuity** separates the upper and lower crust. The **Mohorovičić (Moho) discontinuity (1909)** separates crust and mantle. |
| **Mantle** | Down to about **2,900 km**; about **83% of the Earth's volume**. Its upper part includes the soft, partly molten **asthenosphere** (up to about **400 km**), the main **source of magma** on which plates slide. The **lithosphere** = **crust + rigid top of the mantle** (about **100 km** thick). The **Gutenberg discontinuity** at **2,900 km** separates mantle and core. |
| **Core** | Made mainly of **nickel and iron (NiFe)**. **Outer core** (**2,900–5,150 km**) is **liquid**; its movement creates the **Earth's magnetic field**. **Inner core** (**5,150–6,371 km**) is **solid** because of enormous |

| | |
|---|---|
| | pressure, about as hot as the Sun's surface (**~5,500°C**). The **Lehmann discontinuity** separates them (Inge Lehmann, 1936). **Iron** is the most common element in the whole Earth. |
| **Crust composition** | **Oxygen (~46.6%) > silicon (~27.7%) > aluminium (~8.1%) > iron (~5%)** > calcium > sodium > potassium > magnesium. Aluminium is the most common **metal** in the crust. |

### 3. Rocks

The three rock types

- **Igneous rocks** ('fire rocks') form from cooled **magma or lava**, so they are called **primary rocks**; they have **no fossils**. **Intrusive (plutonic)** rocks cool slowly **inside** the Earth and have **large crystals** (**granite**); **extrusive (volcanic)** rocks cool quickly **on the surface** and have **fine grains** (**basalt**, as in the **Deccan Traps**, which weather into **black cotton soil**).
- **Sedimentary rocks** form from layers of **sediments** pressed together; they cover about **75% of the land surface** but only about **5% of the crust's volume**; they contain **fossils**, coal and oil. Examples: **sandstone, limestone, shale, conglomerate, coal**. **Metamorphic rocks** form when heat and pressure **change** existing rocks: **granite → gneiss, limestone → marble, sandstone → quartzite, shale → slate → schist, coal → graphite**.
- **Andhra Pradesh:** the **Cuddapah basin** has sedimentary rocks such as **limestone** (cement industry) and the **barytes** of **Mangampet**; the **Eastern Ghats** have metamorphic **khondalites** and **charnockites**; **black granite** is quarried at **Chimakurthy** (Prakasam); the **Godavari valley** has **Gondwana** sedimentary rocks.

### 4. Moving plates

- **Continental drift (Alfred Wegener, 1912):** all continents were once one supercontinent, **Pangaea**, surrounded by one ocean, **Panthalassa**; evidence: the **jigsaw fit** of South America and Africa, matching **fossils** and rocks. **Sea-floor spreading (Harry Hess, 1960s):** new crust forms at **mid-ocean ridges**. These led to **plate tectonics (1967)**: the lithosphere is broken into **seven major plates** and many minor ones that move a few cm a year; the **Indian plate** moves north at about **5 cm a year**.
- **Plate boundaries: divergent** (plates move apart; **Mid-Atlantic Ridge**, East African Rift); **convergent** (plates collide; one sinks under the other in **subduction**, forming **deep trenches** such as the **Mariana Trench**, volcanoes and fold mountains; India hitting Eurasia raised the **Himalayas**); **transform** (plates slide past each other; **San Andreas Fault**, California).

### 5. Earthquakes and volcanoes

| Topic | Facts |
|---|---|
| **Earthquake basics** | The point **inside** the Earth where energy is released is the **focus (hypocentre)**; the point on the **surface directly above** it is the **epicentre** (most damage). Recorded by a **seismograph**. **Magnitude** (energy) is measured on the **Richter scale (1935)**, now the **moment magnitude** scale; each step is about **10 times** bigger in shaking and about **32 times** in energy. **Intensity** (damage seen) is measured on the **Modified Mercalli scale (I–XII)**. |

| | |
|---|---|
| **Causes and India** | Plate movement along **faults**, volcanic activity, and human causes such as **large reservoirs** (**Koyna, 1967**) and mining. The **BIS seismic zoning map** (IS 1893) groups India into **zones II to V**; **zone V** (highest risk) covers the **north-east, parts of J&K, Ladakh, Himachal, Uttarakhand, Bihar, the Rann of Kutch and the Andaman and Nicobar Islands** (BIS has been revising this map; check the latest). Major quakes: **Latur (1993)**, **Bhuj (26 January 2001)**, **Sikkim (2011)**. The **Indian Ocean tsunami (26 December 2004)** followed a **magnitude 9.1** quake off **Sumatra**; India's **Tsunami Early Warning Centre** at **INCOIS, Hyderabad** started in **2007**. |
| **Volcanoes** | By activity: **active, dormant, extinct**. By shape: **shield** volcanoes (runny basalt lava, gentle slopes; **Hawaii**); **composite (strato)** volcanoes (layers of lava and ash, **explosive**; **Fuji**, Vesuvius); **calderas** (huge collapsed craters); **fissure** eruptions that pour out **flood basalts** (**Deccan Traps**). About **two-thirds or more** of active volcanoes lie on the **Pacific 'Ring of Fire'**. India's **only active volcano** is **Barren Island** (Andaman Sea); **Narcondam** is **dormant**. |
| **Intrusive forms** | Magma that cools **inside** the crust forms **batholiths** (huge granite masses), **laccoliths** (dome-shaped), **lopoliths** (saucer-shaped), **sills** (horizontal sheets) and **dykes** (vertical walls). |

**EXAM TRAP**

Kola **~12 km**; centre **6,371 km**; **P** fastest, all media; **S** solids only; surface waves slowest, most damage; shadow zone **105°–145°**; no S beyond 105° = **liquid outer core**; crust ocean **~5 km**, continent **~30–35 km**; **SiAl, SiMa**; **Conrad**; **Moho (1909)**; mantle to **2,900 km** (**~83%** volume); **asthenosphere** (magma); lithosphere = crust + top mantle; **Gutenberg 2,900 km**; outer core **liquid** (magnetism); inner core **solid**; **Lehmann 5,150 km**; **NiFe**; crust **O > Si > Al > Fe**; igneous (granite, basalt, no fossils); sedimentary (**75%** surface, fossils); metamorphic (gneiss, **marble**, quartzite, slate); AP: **Cuddapah, khondalite, charnockite, Chimakurthy**; **Wegener 1912 (Pangaea)**; **Hess**; **plate tectonics 1967**; divergent, convergent, transform; **Richter 1935**, **Mercalli I–XII**; zones **II–V**; **Koyna 1967, Latur 1993, Bhuj 2001**; tsunami **2004**, **INCOIS 2007**; shield, composite, caldera, fissure; **Ring of Fire**; **Barren Island** (active), **Narcondam** (dormant); batholith, laccolith, lopolith, sill, dyke.

## Test yourself: 5 questions (tap an option to see the answer)

**PRACTICE**

**Consider the following statements:**

**1. P waves can travel through solids, liquids and gases.**

**2. S waves cannot travel through liquids.**

**3. Surface waves are the fastest seismic waves.**

**4. The absence of S waves beyond about 105° from the epicentre shows that the outer core is liquid.**

**Which of the above are correct?**

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 4 only

**Answer: D.** Surface waves are the slowest, though the most destructive; P waves are the fastest.

**PRACTICE**

**Match List-I (discontinuity) with List-II (layers it separates):**

| List-I (discontinuity) | List-II (layers it separates) |
|---|---|
| **A.** Mohorovičić | **1.** Upper and lower crust |
| **B.** Gutenberg | **2.** Outer and inner core |
| **C.** Lehmann | **3.** Mantle and core |
| **D.** Conrad | **4.** Crust and mantle |

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 3 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

**Answer: (2)** Moho: crust–mantle; Gutenberg: mantle–core; Lehmann: outer–inner core; Conrad: within the crust.

**PRACTICE**

**Which one of the following statements is NOT correct?**

A. Marble is formed by the metamorphism of sandstone.

B. Barren Island is India's only active volcano.

C. The Richter scale measures the magnitude of an earthquake.

D. The Mercalli scale measures the intensity of an earthquake.

**Answer: A.** Marble comes from limestone; sandstone becomes quartzite.

**PRACTICE**

**Assertion (A): Sedimentary rocks often contain fossils.**
**Reason (R): Sedimentary rocks are formed by the cooling and solidification of magma.**
**Choose the correct answer:**

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

**Answer: C.** A is true. R is false: sedimentary rocks form from layers of sediment, which can bury remains of plants and animals; igneous rocks form from magma.

**PRACTICE**

**A seismograph station about 160° away from an earthquake records P waves but no S waves. What does this best show?**

A. The earthquake was very shallow

B. The waves passed through the liquid outer core, which stops S waves

C. The station lies at the epicentre

D. The seismograph was faulty

**Answer: B.** S waves cannot cross the liquid outer core, so stations beyond about 105° get no S waves.

**RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS**

1. Write the sources, the three wave types and the shadow zone. 2. Write the layers with depths, discontinuities and composition. 3. Write the rock types with examples, plate tectonics and boundaries, earthquake terms and scales, and volcano types.

## POLITY · The 74th Amendment (urban local government) and its implementation

**UNDERSTAND IT FIRST**

**What is it?** Towns and cities are run by **urban local bodies (ULBs)**, called **municipalities** in common speech. They supply **water, drains, roads, street lights, waste collection, birth and death registration** and more. Before 1992 many State Governments **kept them under their thumb**: elections were postponed for years and bodies were **superseded**. The **74th Amendment (1992)** did for towns what the 73rd did for

villages: it gave ULBs **constitutional status**, **regular elections**, **reservation** and a **finance commission**. Learn it as **history → the Amendment's features → planning committees → Twelfth Schedule → other urban bodies → problems and schemes**.

## 1. Background

- The first municipal corporation in India was set up at **Madras in 1688**; Bombay and Calcutta followed in **1726**. **Lord Ripon's Resolution of 1882** is called the **'Magna Carta of local self-government'**, and Ripon the **father of local self-government** in India. Local government became a **provincial (State) subject** (Government of India Act 1919, dyarchy) and is in the **State List (Entry 5)** today.
- The **65th Amendment Bill (1989)** on municipalities failed in the Rajya Sabha. The **74th Amendment Act** was passed in **1992** and came into force on **1 June 1993**. It added **Part IXA** (**Articles 243P to 243ZG**) and the **Twelfth Schedule** (**18 subjects**).

## 2. Main features

Urban local bodies under the 74th Amendment

| Article | Provision |
|---|---|
| **243Q** | **Three types**: **Nagar Panchayat** (area in **transition** from rural to urban), **Municipal Council** (**smaller** urban area), **Municipal Corporation** (**larger** urban area). An **industrial township** run by an industry may be excluded. |
| **243R, 243S** | Members **directly elected** from **wards**; the State may add **nominated experts** (**without voting rights**), MPs and MLAs. **Wards Committees** are compulsory where a municipality's population is **3 lakh or more**. |
| **243T, 243U, 243V** | Reservation for **SCs and STs** by population and **not less than one-third for women**; reservation of **chairperson** posts as State law provides; OBC reservation **optional**. **Five-year term**; after dissolution, **election within six months**. Minimum age **21**. |
| **243W, 243X** | State may give powers over **Twelfth Schedule** subjects; may allow ULBs to levy **taxes, duties, tolls and fees** and give **grants**. |
| **243Y, 243Z, 243ZA** | The **State Finance Commission** (set up under **243I**) also reviews **municipal finances**; **audit** by State law; elections by the **State Election Commission**. |
| **243ZD** | **District Planning Committee (DPC)** in every district to **combine the plans of panchayats and municipalities** into a **draft district plan**; at least **four-fifths** of members are elected by elected members of the district panchayat and municipalities, in proportion to **rural and urban population**. |
| **243ZE** | **Metropolitan Planning Committee (MPC)** for every **metropolitan area** (population **10 lakh or more**); at least **two-thirds** of members elected by elected members of the municipalities and panchayats in the area. |

| | |
|---|---|
| **243ZC, 243ZG** | Does **not apply to Scheduled Areas**; Parliament may extend it (no such law yet). Courts **cannot interfere** in municipal **electoral matters** except by an election petition. |

**Twelfth Schedule (18 subjects):** urban and town planning; land use and buildings; economic and social development planning; roads and bridges; water supply; public health, sanitation and solid waste; fire services; urban forestry and environment; weaker sections and the disabled; slum improvement; urban poverty alleviation; parks and playgrounds; cultural, educational and aesthetic activities; burial and cremation grounds; cattle pounds and cruelty to animals; vital statistics (births and deaths); street lighting, parking and bus stops; slaughterhouses and tanneries.

### 3. Other urban bodies and implementation

- **Other urban bodies: Cantonment Boards** (for military areas, under the **Union Ministry of Defence**, **Cantonments Act 2006**); **township** administrations of public-sector industries; **port trusts**; **notified area committees**; and **special-purpose agencies** (development authorities, water boards), often called **parastatals**.
- **Problems:** ULBs raise little **own revenue** (property tax is under-collected) and depend on **grants**; many Twelfth Schedule functions remain with **State departments and parastatals**; **elections are often delayed** (the Supreme Court in **Suresh Mahajan (2022)** said they cannot be postponed); **Mayors** are mostly **indirectly elected** with **short terms** and little power, while the **commissioner** (a State officer) runs the administration; DPCs and MPCs are **weak or absent** in many States; staff and planning **capacity** are low. The **Second ARC (6th Report, 2007)** suggested stronger Mayors and more devolution.
- **Schemes: JNNURM (2005)**; **Swachh Bharat Mission–Urban (2014; 2.0 in 2021)**; **Smart Cities Mission (2015, 100 cities;** in AP: **Visakhapatnam, Kakinada, Tirupati, Amaravati)**; **AMRUT (2015; 2.0 in 2021)** for water and sewerage; **PMAY–Urban (2015; 2.0 in 2024)**; **municipal bonds**; **Finance Commission grants** tied to property tax reforms. **AP: AP Municipalities Act, 1965** and **AP Municipal Corporations Act, 1994**; **ward secretariats** (2019) bring services closer to people.

**EXAM TRAP**

**Madras 1688**; Bombay, Calcutta **1726**; **Ripon 1882** (Magna Carta); State List **Entry 5**; **65th Bill 1989 failed**; **74th: 1992, in force 1 June 1993**; **Part IXA, 243P–243ZG**; **Twelfth Schedule, 18**; **243Q**: Nagar Panchayat (transition), Council (smaller), Corporation (larger); nominated members **no vote**; **Wards Committees ≥3 lakh (243S)**; **243T** SC/ST, women ≥1/3, OBC optional; **243U** 5 years, 6 months; age **21**; **243Y** SFC; **243ZA** SEC; **243ZD DPC (4/5 elected)**; **243ZE MPC (10 lakh+, 2/3 elected)**; **243ZC** not for Scheduled Areas; Cantonment Boards (**Defence, 2006 Act**); parastatals; problems: revenue, delayed polls (**Suresh Mahajan 2022**), weak Mayor; **2nd ARC 6th report**; **JNNURM 2005, SBM-U 2014, Smart Cities 2015 (AP 4), AMRUT 2015, PMAY-U 2015**; AP Acts **1965, 1994**.

**Test yourself: 5 questions (tap an option to see the answer)**

**PRACTICE**

**Consider the following statements about the 74th Constitutional Amendment:**

**1. It added Part IXA and the Twelfth Schedule to the Constitution.**

**2. The Twelfth Schedule contains 18 subjects.**

**3. Wards Committees are required in municipalities with a population of three lakh or more.**

**4. A Metropolitan Planning Committee is required for every area with a population of five lakh or more.**

**Which of the above are correct?**

A. 1, 2 and 3 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

**Answer: A.** A metropolitan area means a population of ten lakh or more.

**PRACTICE**

**Match List-I (urban body) with List-II (feature):**

| List-I (urban body) | List-II (feature) |
|---|---|
| A. Nagar Panchayat | 1. Under the Union Ministry of Defence |
| B. Municipal Council | 2. For a larger urban area |
| C. Municipal Corporation | 3. For a smaller urban area |
| D. Cantonment Board | 4. For an area changing from rural to urban |

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

**Answer: (3)** Nagar Panchayat: transitional; Council: smaller; Corporation: larger; Cantonment: Defence.

**PRACTICE**

**Which one of the following statements is NOT correct?**

A. The 74th Amendment came into force on 1 June 1993.

B. The District Planning Committee combines the plans of panchayats and municipalities.

C. Lord Ripon's Resolution of 1882 is called the Magna Carta of local self-government.

D. The 74th Amendment requires the Mayor of every municipal corporation to be directly elected.

**Answer: D.** The method of electing chairpersons and Mayors is left to State law.

**PRACTICE**

**Assertion (A): Many urban local bodies in India have weak finances.**
**Reason (R): The Twelfth Schedule lists 18 functions that may be given to municipalities.**
**Choose the correct answer:**

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

**Answer: B.** Both are true, but the list of functions does not explain weak finances; low own revenue, poor property tax collection and dependence on grants do.

**PRACTICE**

**A city region of 15 lakh people includes a municipal corporation, three municipalities and several gram panchayats. Under the Constitution, which body should prepare a development plan for the whole region?**

A. The Wards Committee

B. The State Finance Commission

C. The Metropolitan Planning Committee

D. The State Election Commission

**Answer: C.** Art 243ZE provides a Metropolitan Planning Committee for areas of ten lakh or more.

**RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS**

1. Write the history and dates. 2. Write the Article-wise table from 243Q to 243ZG and the 18 subjects. 3. Write other urban bodies, the problems and the schemes, with AP's laws.

## SCIENCE · Biodiversity: hotspots, loss and conservation

**UNDERSTAND IT FIRST**

**What is it? Biodiversity** (biological diversity) is the **variety of life**: all the different **genes, species and ecosystems** on Earth. It gives us **food, medicines, clean water, pollination, fertile soil** and a stable climate. But species are now disappearing **far faster than normal**, mostly because of humans. India is one of the world's **richest** countries in biodiversity, so the topic is asked often, along with **international agreements and Indian laws**. Learn it as **meaning and levels → India's richness → hotspots → causes of loss → conservation methods → treaties and laws → AP's special species**. (Protected areas, CITES and the IUCN Red List come on Day 27.)

### 1. Meaning, levels and India's richness

- The term **'biodiversity'** was popularised by **E.O. Wilson** (from a 1986 forum; the word was coined by **Walter G. Rosen**). **Three levels: genetic diversity** (variation within a species, e.g. thousands of **rice** varieties), **species diversity** (number and variety of species) and **ecosystem diversity** (forests, grasslands, wetlands, deserts, coral reefs). **Alpha** diversity is within one community, **beta** between communities, and **gamma** across a whole region.
- **India** is one of the **17 megadiverse countries**: with only about **2.4% of the world's land**, it has about **7–8% of recorded species**, including over **45,000 plant species** and over **1 lakh animal species**. It has **10 biogeographic zones** (from the trans-Himalaya to the islands), and is a centre of origin of crops like **rice, sugarcane, mango and pepper**.

### 2. Biodiversity hotspots

India's four biodiversity hotspots

- The idea was given by **Norman Myers (1988)** and is used by **Conservation International**. A region is a **hotspot** if it has **at least 1,500 endemic vascular plant species** (found nowhere else) and has **lost at least 70%** of its original natural vegetation. There are **36 hotspots** worldwide; they cover only about **2.5% of the land** but hold over **half of the world's endemic plant species**.
- **India has parts of four:** the **Himalaya**, **Indo-Burma** (north-east), the **Western Ghats and Sri Lanka**, and **Sundaland** (**Nicobar Islands**). **Exam trap:** the **Eastern Ghats are not a hotspot**, although they have many endemic species.

### 3. Loss of biodiversity

- The **IPBES report (2019)** warned that about **one million species** face extinction. Main causes are remembered as **'HIPPO'**: **Habitat loss** and fragmentation (the biggest cause), **Invasive alien species**,

**Pollution**, human **Population** growth and **Overexploitation** (over-hunting, over-fishing, illegal trade), plus **climate change**. When one species dies out, others that depend on it may also be lost (**co-extinction**).

- **Invasive species in India: Lantana**, **Parthenium** ('congress grass'), **water hyacinth** (Eichhornia) choking lakes, **Prosopis juliflora** (sarkaru tumma) and the **African catfish** in ponds. **Lost species:** the **cheetah** was declared **extinct in India in 1952**; **Project Cheetah** brought cheetahs to **Kuno (MP)** in **September 2022**.

### 4. Conservation, treaties and laws

| Item | Details |
|---|---|
| **In-situ** | Protecting species **in their natural home**: national parks, wildlife sanctuaries, **biosphere reserves**, community reserves and **sacred groves** (e.g. **kavu** in Kerala, **Mawphlang** in Meghalaya, **orans** in Rajasthan). Best for whole ecosystems. |
| **Ex-situ** | Protecting species **outside** their natural home: **zoos**, **botanical gardens**, **seed and gene banks** (**NBPGR, New Delhi**; the **Svalbard Global Seed Vault**, Norway, 2008), **cryopreservation** in liquid nitrogen at **−196°C**, **tissue culture** and **captive breeding** (e.g. vultures). |
| **Convention on Biological Diversity (CBD)** | Signed at the **Rio Earth Summit (1992)**, in force **1993**; three goals: **conservation, sustainable use, and fair and equitable sharing of benefits** from genetic resources. **Cartagena Protocol (2000)** on **biosafety** of living modified organisms; **Nagoya Protocol (2010, in force 2014)** on **access and benefit-sharing (ABS)**; **Kunming–Montreal Global Biodiversity Framework (COP15, December 2022)**: **23 targets** for 2030, including **'30 by 30'** (protect **30% of land and sea by 2030**); **COP16 (Cali, 2024)** set up the **Cali Fund** for benefits from digital sequence information. **22 May** is International Day for Biological Diversity. |
| **Biological Diversity Act, 2002** | India's law to carry out the CBD, with a **three-tier** structure: **National Biodiversity Authority (NBA, Chennai, 2003)**, **State Biodiversity Boards** and **Biodiversity Management Committees (BMCs)** in local bodies, which keep **People's Biodiversity Registers (PBRs)**. Foreign users need **NBA approval**; Indian companies must **inform the State Board**; **benefits** must be shared with local communities. The **2023 Amendment** eased rules for **AYUSH** practitioners and **decriminalised** many offences. |

**Andhra Pradesh's special species: red sanders** (**Pterocarpus santalinus**), **endemic** to the southern **Eastern Ghats** (**Seshachalam, Lankamala, Palakonda**); **Jerdon's courser**, rediscovered in **1986** near **Sri Lankamalleswara** (Kadapa); the **golden gecko** of the **Tirumala** hills; the **Great Indian Bustard** at **Rollapadu**; the **fishing cat** (AP's **State animal is the blackbuck**) in the **Coringa and Krishna** mangroves.

**EXAM TRAP**

Word coined by **Rosen**, popularised by **E.O. Wilson**; **genetic, species, ecosystem**; alpha, beta, gamma; India **megadiverse (1 of 17)**, **2.4% land, 7–8% species**, **10 biogeographic zones**; hotspot: **Myers 1988**, **≥1,500 endemic plants**, **≥70% lost**; **36** hotspots, **2.5% land**; India: **Himalaya, Indo-Burma, Western Ghats–Sri Lanka, Sundaland (Nicobar)**; **Eastern Ghats not a hotspot**; **IPBES 2019 (1 million)**; **HIPPO** + climate; invasives: **Lantana, Parthenium, water hyacinth, Prosopis, African catfish**; cheetah extinct **1952**, **Kuno 2022**; in-situ (PAs, sacred groves); ex-situ (zoos, **NBPGR**, **Svalbard 2008**, **−196°C**); **CBD 1992/1993** (3

goals); **Cartagena 2000 (biosafety)**; **Nagoya 2010/2014 (ABS)**; **KMGBF 2022 (23 targets, 30×30)**; **Cali Fund 2024**; **22 May**; **BD Act 2002: NBA Chennai 2003, SBB, BMC (PBR)**; **2023 amendment**; AP: **red sanders, Jerdon's courser 1986, golden gecko, GIB Rollapadu, fishing cat**.

**Test yourself: 5 questions (tap an option to see the answer)**

PRACTICE

**Consider the following statements about biodiversity hotspots:**

**1. India has parts of four of the world's 36 biodiversity hotspots.**

**2. The Eastern Ghats are one of these hotspots.**

**3. A hotspot must have at least 1,500 endemic vascular plant species.**

**4. A hotspot must have lost at least 70% of its original natural vegetation.**

**Which of the above are correct?**

A. 1 and 3 only

B. 1, 3 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

**Answer: B.** India's four are the Himalaya, Indo-Burma, Western Ghats–Sri Lanka and Sundaland; the Eastern Ghats are not a hotspot.

PRACTICE

**Match List-I (agreement or law) with List-II (subject):**

| List-I (agreement or law) | List-II (subject) |
|---|---|
| A. Cartagena Protocol | 1. 2002; National Biodiversity Authority |
| B. Nagoya Protocol | 2. '30 by 30' target, 2022 |
| C. Kunming–Montreal Framework | 3. Access and benefit-sharing |
| D. Biological Diversity Act | 4. Biosafety of living modified organisms |

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 | B → 3 | C → 2 | D → 1

**Answer: (4)** Cartagena: biosafety; Nagoya: ABS; Kunming–Montreal: 30×30; BD Act: NBA.

**PRACTICE**

**Which one of the following pairs is NOT correctly matched?**

A. Lantana : invasive weed

B. Parthenium : 'congress grass'

C. Red sanders : endemic to the Western Ghats

D. Svalbard : global seed vault

**Answer: C.** Red sanders is endemic to the southern Eastern Ghats (Seshachalam, Lankamala, Palakonda) in Andhra Pradesh.

**PRACTICE**

**Assertion (A): Species in biodiversity hotspots face a high risk of extinction.**
**Reason (R): Hotspots have many endemic species and have already lost most of their original habitat.**
**Choose the correct answer:**

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

**Answer: A.** Species found nowhere else, living in shrinking habitat, are the most likely to be lost.

**PRACTICE**

**A foreign company wants to use a medicinal plant traditionally used by a tribal community in Andhra Pradesh to make a commercial drug. Under Indian law, what must it do?**

A. Nothing, as plants are free for anyone to use

B. Get approval from the National Biodiversity Authority and share benefits with the local community

C. Only get a patent from the Patent Office

D. Only get a licence from the Forest Department

**Answer: B.** Under the Biological Diversity Act, 2002 (following the Nagoya Protocol), foreign users need NBA approval, and benefits must be shared with the community.

**RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS**

1. Write the meaning, three levels and India's figures. 2. Write the hotspot criteria and India's four, then HIPPO and invasive species. 3. Write in-situ vs ex-situ, the CBD and its protocols, the BD Act's three tiers and AP's special species.

## NOTEBOOK PAGE · Day 26: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

### GEOGRAPHY: Interior of the Earth

| Area | Write |
|---|---|
| Interior | Sources; P, S and surface waves; shadow zone; layers with depths and discontinuities; composition |
| Rocks | Three types, examples, changes in metamorphism; AP rocks |
| Plates, earthquakes, volcanoes | Wegener, Hess, three boundaries; focus, epicentre, scales, zones, big quakes; volcano types, Ring of Fire, Barren Island, intrusive forms |

### POLITY: 74th Amendment

- History (1688, Ripon 1882); dates; Part IXA, Articles 243P–243ZG; three bodies; wards committees; reservation; term; SFC, SEC; DPC and MPC; Twelfth Schedule (18); other urban bodies; problems; schemes; AP's laws and bodies.

### SCIENCE: Biodiversity

Write: meaning and three levels; India's share; hotspot criteria and India's four; causes of loss (HIPPO) and invasive species; in-situ vs ex-situ; CBD and its protocols, Kunming–Montreal; Biological Diversity Act 2002 and three tiers; AP's special species.